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**DATABASE DESIGN AND IMPLEMENTATION: ROAD SIGN AND ROAD
STATE MOBILE NOTIFICATION APPLICATION**

GROUP 5

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ABSTRACT

This report presents the comprehensive backend system design and implementation for WayWatch, a mobile road management application that provides real-time road condition monitoring, incident reporting, and location-based services. The system architecture encompasses a robust PostgreSQL database with PostGIS geospatial capabilities and a comprehensive API implementation supporting user authentication, real-time location tracking, crowd-sourced incident reporting, and automated alert distribution.

The database design follows established software engineering principles with 12 core tables managing users, preferences, locations, road signs, conditions, alerts, and navigation data. The API implementation provides seamless integration between mobile clients and backend services, supporting features including user registration with OTP verification, continuous GPS location updates, dynamic road sign retrieval, crowd reporting mechanisms, and proximity-based alert broadcasting.

Performance optimization techniques include spatial indexing for geolocation queries, role-based access control for security, and scalable alert distribution systems capable of notifying users within specified geographical radii. The system demonstrates successful integration of modern web technologies with geospatial data management for real-time road safety applications.

1. INTRODUCTION

Road safety and traffic management represent critical challenges in modern urban infrastructure, with traditional static signage and delayed incident reporting contributing to accidents and traffic congestion. WayWatch addresses these challenges through a comprehensive mobile application that leverages real-time data collection, crowd-sourced reporting, and location-based services to enhance road safety and traffic awareness.

The backend system serves as the foundation for WayWatch's functionality, providing secure data management, real-time processing capabilities, and scalable service delivery. This report documents the complete backend architecture, encompassing both database design and API implementation phases of the development process.

The system requirements include support for multiple user types (Drivers, Pedestrians, State Agencies), real-time GPS tracking, dynamic content delivery based on location, crowd-sourced incident reporting, and automated alert distribution. The backend must handle concurrent users, process geospatial queries efficiently, and maintain data integrity while supporting real-time operations.

This implementation utilizes PostgreSQL with PostGIS extension for geospatial data management, Node.js/Express.js for API development, and modern security practices including JWT authentication and encrypted data transmission. The architecture supports scalable deployment on cloud platforms with provisions for high availability and disaster recovery.

2. SYSTEM OVERVIEW

WayWatch is designed as a comprehensive road management solution that connects users, road infrastructure data, and real-time incident information through a centralized backend system. The application serves three primary user categories: drivers seeking navigation assistance and hazard awareness, pedestrians requiring road safety information, and state agencies managing road infrastructure and monitoring traffic conditions.

The backend system processes continuous streams of location data from mobile clients, matches user positions with relevant road infrastructure information, facilitates crowd-sourced incident reporting, and distributes contextual alerts to users within affected geographical areas. This real-time processing capability requires robust database design, efficient API endpoints, and scalable alert distribution mechanisms.

The system architecture emphasizes data accuracy, real-time responsiveness, and user safety, implementing features such as OTP-verified user registration, encrypted password storage,

JWT-based session management, and proximity-based alert broadcasting with 1-2 kilometer radius coverage for incident notifications.

3. DATABASE DESIGN AND IMPLEMENTATION

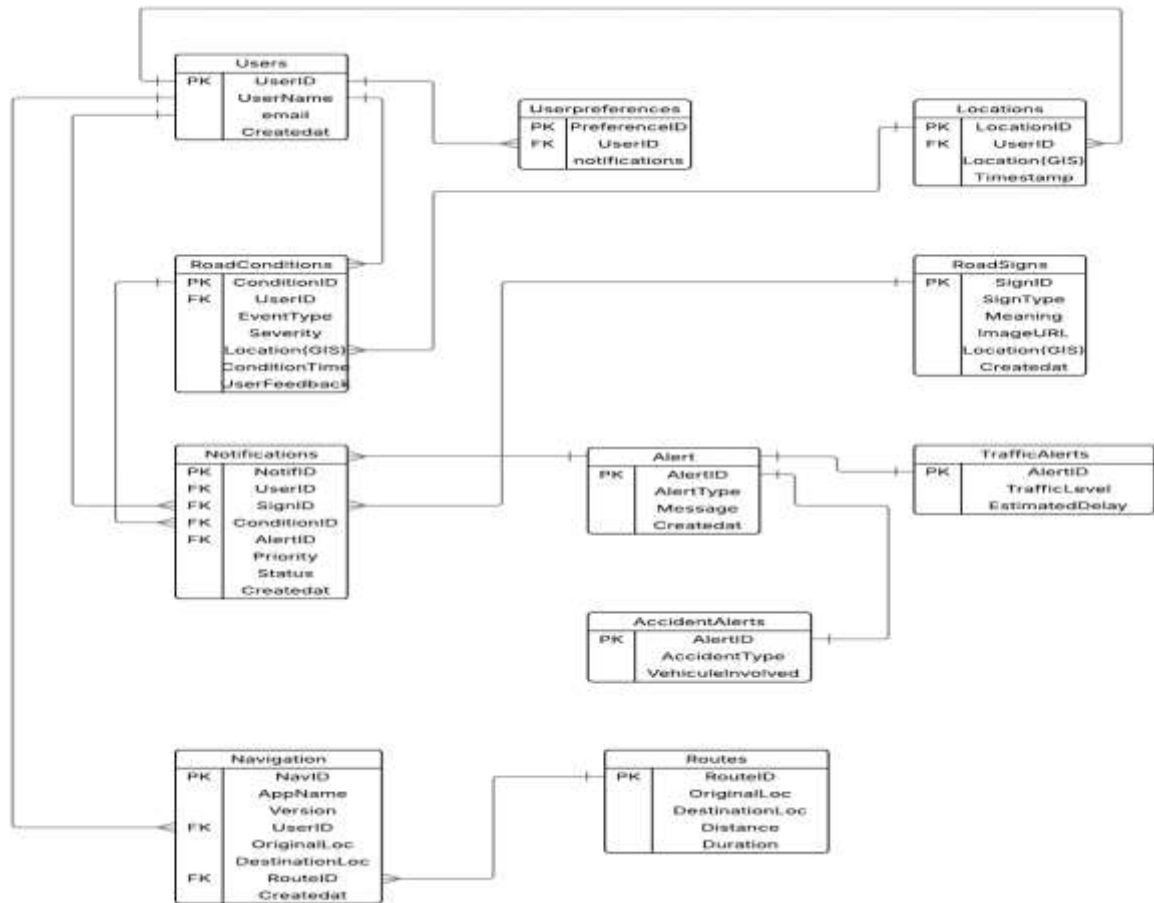
3.1 Database Architecture

The WayWatch database architecture is built on PostgreSQL 17+ with PostGIS extension, providing robust relational data management combined with advanced geospatial capabilities. The choice of PostgreSQL ensures compatibility with Node.js/Express.js backend implementation while supporting complex spatial queries required for location-based services.

Key Architectural Decisions:

- **PostgreSQL with PostGIS:** Enables efficient storage and querying of geographical data
- **UUID Primary Keys:** Ensures globally unique identifiers across distributed systems
- **GEOMETRY(Point, 4326):** Standardized GPS coordinate storage using WGS84 coordinate system
- **JSONB Data Types:** Flexible storage for route information and complex data structures
- **Spatial Indexing:** GIST indexes optimize location-based query performance

Entity-Relationship Diagram for a Road Traffic and Alert Management System



3.2 Schema Design

The database schema comprises 12 core tables designed following UML Class Diagrams and Data Flow Diagrams, ensuring normalized data structure and efficient query patterns:

Core Tables Overview

Table	Purpose
users	Stores user profiles and roles (Driver, Pedestrian, StateAgency)
user_preferences	Stores notification settings and user preferences
locations	Stores geospatial location points from GPS tracking
road_signs	Stores road sign data with geographical positioning
road_conditions	Stores real-time or reported road hazards and conditions
alerts	Base table for alert messages with inheritance structure
traffic_alerts	Specialized alerts for traffic-related incidents
accident_alerts	Specialized alerts for accident reporting
notifications	Manages alert delivery queue and status tracking

routes	Stores routing information including distance and duration
navigation	Tracks individual navigation sessions with route details

3.3 Table Relationships

The database implements a comprehensive relationship structure ensuring data integrity and supporting complex queries:

- **users → user_preferences, locations, notifications, navigation:** One-to-Many relationships supporting user-centric data organization
- **alerts → traffic_alerts, accident_alerts:** One-to-One inheritance relationships enabling specialized alert types
- **notifications → users, alerts, road_signs, road_conditions:** Many-to-One relationships supporting flexible notification routing
- **road_conditions → users, locations:** Many-to-One relationships linking incidents to reporters and locations
- **navigation → routes, users:** Many-to-One relationships tracking user navigation sessions

All foreign key relationships implement appropriate cascading behavior with ON DELETE CASCADE for dependent data and ON DELETE SET NULL for reference preservation.

3.4 Implementation Details

Performance Optimization:

- Spatial Indexes for Location-Based Queries

```
CREATE INDEX idx_locations_geom ON locations USING GIST(location);
```

```
CREATE INDEX idx_road_signs_geom ON road_signs USING GIST(location);
```

```
CREATE INDEX idx_road_conditions_geom ON road_conditions USING GIST(location);
```

- Standard Indexes for Foreign Keys

```
CREATE INDEX idx_user_preferences_user_id ON user_preferences(user_id);
```

```
CREATE INDEX idx_notifications_user_id ON notifications(user_id);
```


Data Types and Constraints:

- UUID primary keys ensure global uniqueness
- CHECK constraints validate data ranges and enumerated values
- ARRAY data types support flexible preference storage
- GEOMETRY types with SRID 4326 ensure consistent GPS coordinate handling

3.5 Security Measures

Access Control:

- Role-based access control (RBAC) with specific database roles (admin, readonly, frontend)
- Restricted production database access limited to backend service accounts
- SSL/TLS encryption for all database communications

Data Validation:

- Comprehensive constraint enforcement using CHECK, UNIQUE, NOT NULL, and foreign key constraints
- Parameterized query preparation to prevent SQL injection attacks
- Input validation at application layer before database interaction

Data Protection:

- Encryption at rest through PostgreSQL native capabilities
- Secure password hashing using industry-standard algorithms
- Audit logging for sensitive operations

3.6 Reliability Measures

Data Integrity:

- Foreign key constraints with appropriate cascading rules
- Transaction-based operations ensuring atomicity
- Regular constraint validation and consistency checking

Backup and Recovery:

- Automated database dumps with configurable retention policies
- Write-Ahead Logging (WAL) archiving for point-in-time recovery
- Replication support for high availability deployments

Performance Monitoring:

- Query performance analysis and optimization
- Spatial index maintenance and statistics updates
- Connection pooling and resource management

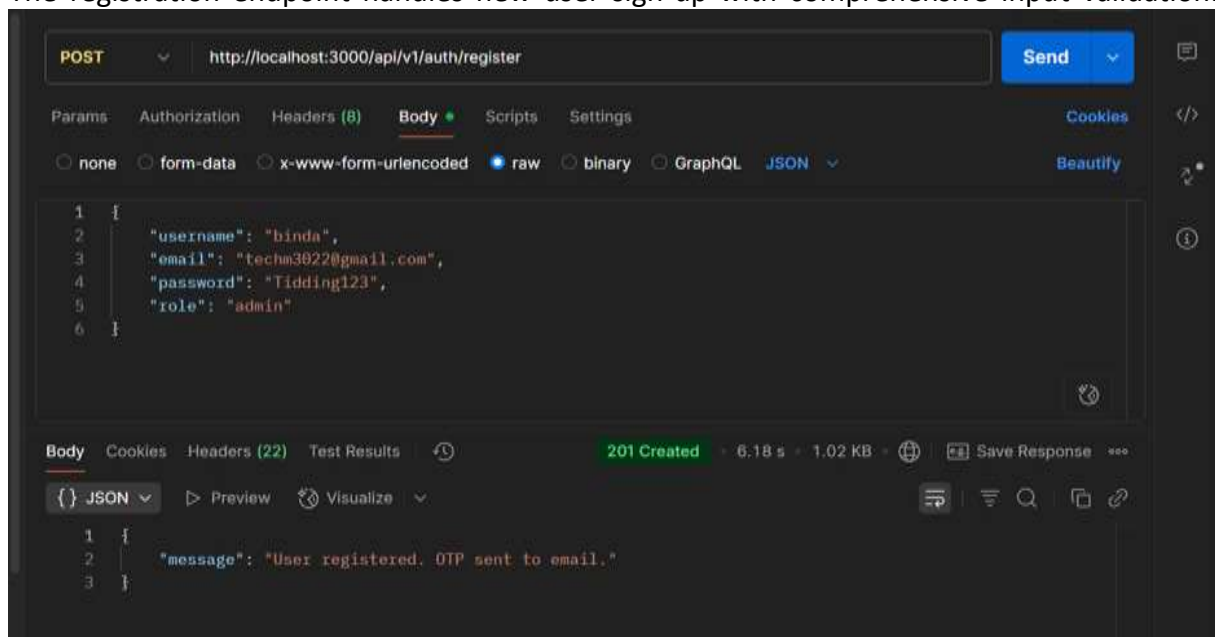
4. BACKEND API IMPLEMENTATION

4.1 User Authentication System

The authentication system provides secure user onboarding and session management through a multi-step verification process.

Registration Screen Implementation:

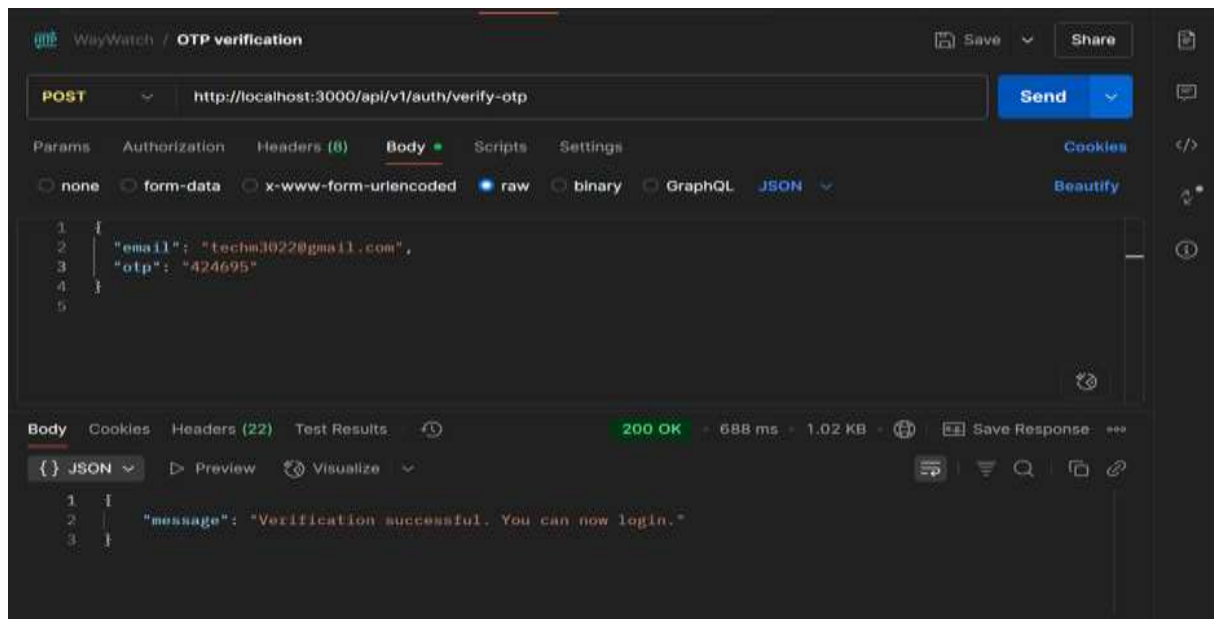
The registration endpoint handles new user sign-up with comprehensive input validation:



- **Input Processing:** Captures username, email, and password with client-side and server-side validation
- **Password Security:** Implements bcrypt hashing with configurable salt rounds
- **Database Integration:** Stores user information with UUID generation and constraint validation
- **Response Handling:** Returns success confirmation or detailed error messages

4.2 OTP Verification

OTP Verification Screen:



The OTP verification process ensures email authenticity and account activation:

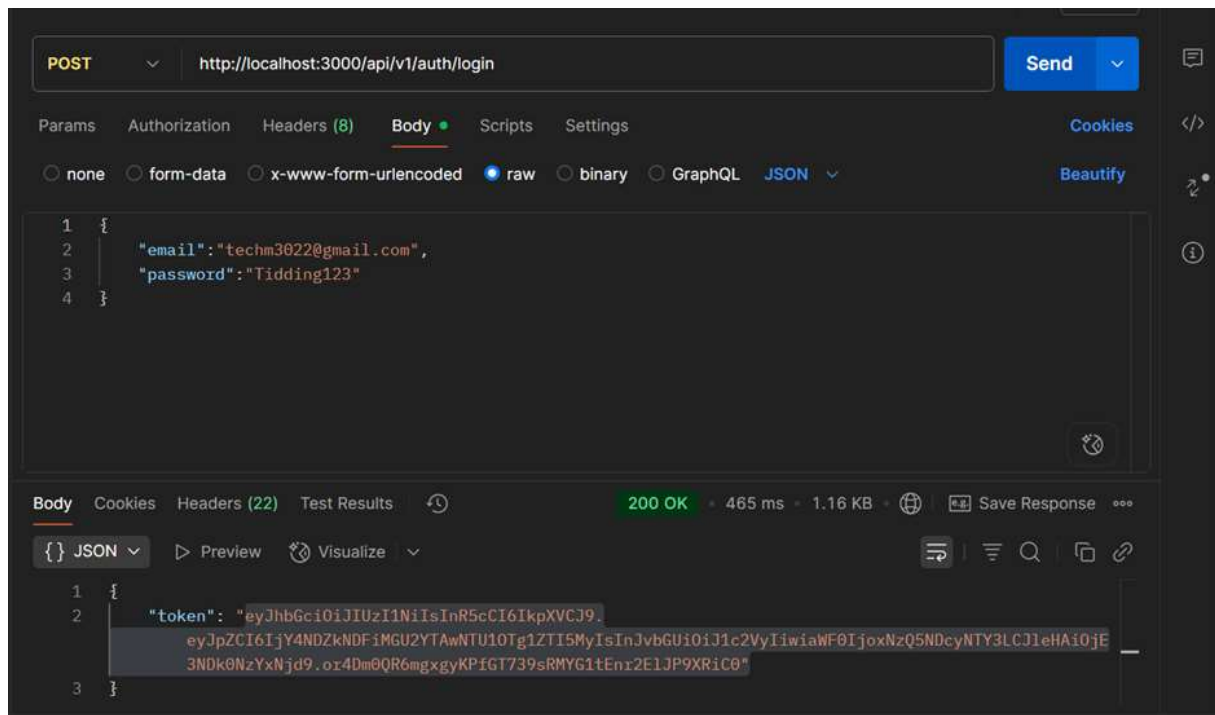
- **OTP Generation:** Creates time-limited verification codes using cryptographic random generation
- **Email Delivery:** Integrates with email service providers for reliable OTP distribution
- **Verification Logic:** Compares submitted OTP with stored values including expiration validation
- **Account Activation:** Updates user status upon successful verification

Implementation Features:

- Configurable OTP expiration time (default: 10 minutes)
- Rate limiting to prevent abuse
- Automatic cleanup of expired OTP records
- Resend functionality with cooldown periods

4.3 Login Authentication

Login Screen Implementation:



The login system provides secure credential verification and session establishment:

- **Credential Validation:** Verifies email and password against stored hash values
- **JWT Token Generation:** Creates signed tokens with user identification and expiration
- **Session Management:** Implements refresh token rotation for extended sessions
- **Security Headers:** Sets appropriate security headers and cookie configurations

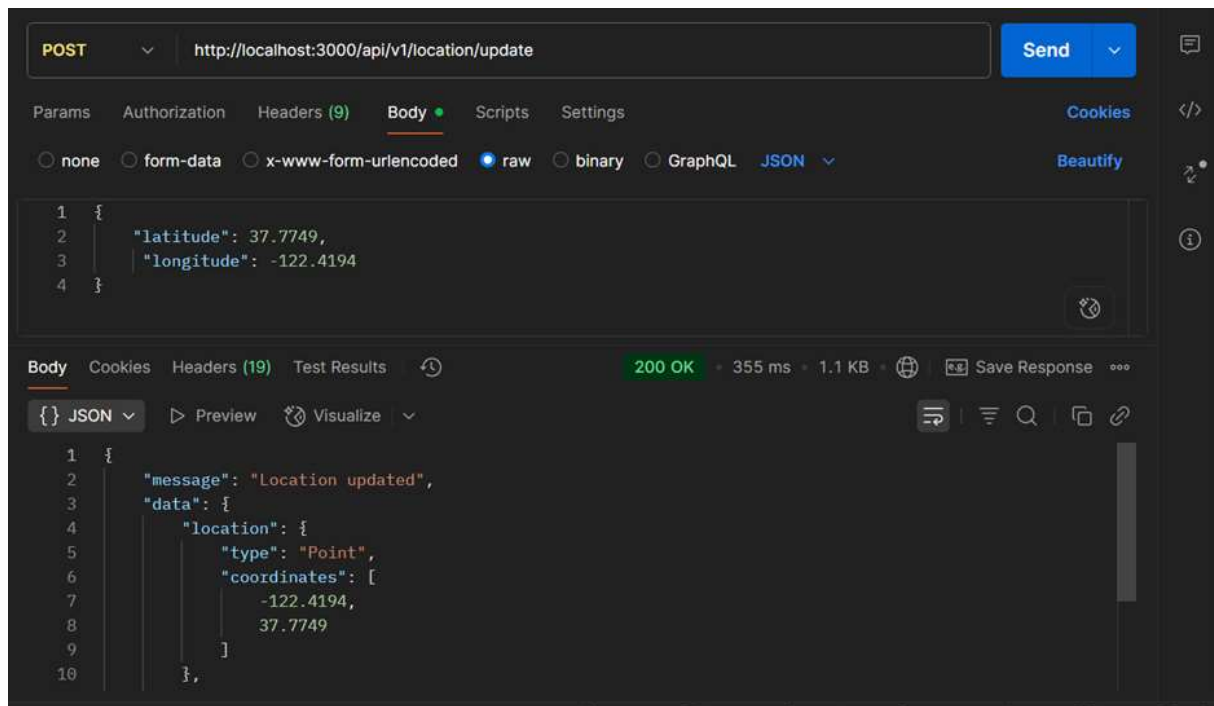
JWT Token Structure:

- **Header:** Algorithm and token type specification
- **Payload:** User ID, role, and expiration timestamp
- **Signature:** HMAC-SHA256 signature for integrity verification

4.4 Location Services

Real-Time Location Tracking

Location Update Screen:



The location service manages continuous GPS tracking with optimized data transmission:

- **GPS Data Collection:** Captures latitude, longitude, altitude, and accuracy metrics
- **Transmission Optimization:** Implements intelligent batching and compression for bandwidth efficiency
- **Database Storage:** Stores location points with PostGIS geometry types and temporal indexing
- **Privacy Controls:** Provides user-configurable location sharing preferences

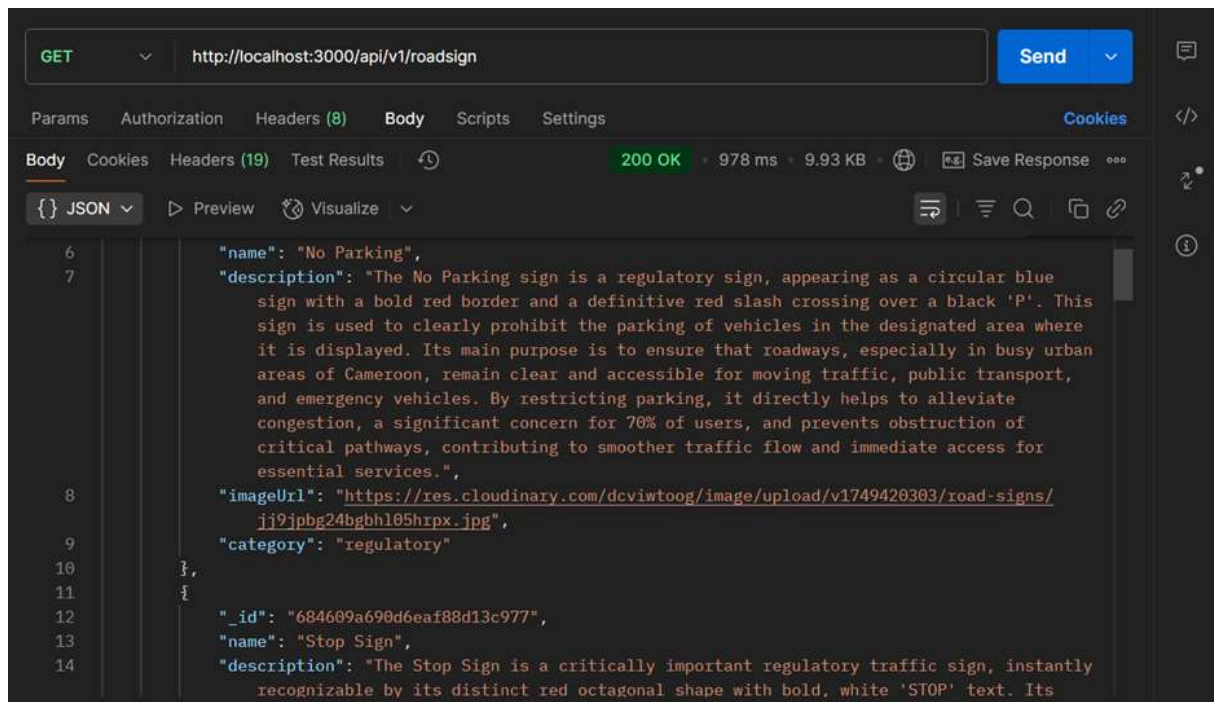
Performance Considerations:

- Configurable update intervals based on user preferences and battery optimization
- Geofencing capabilities to trigger location-based events
- Historical location data retention policies
- Spatial clustering algorithms for privacy-preserving location aggregation

4.5 Road Sign Management

Dynamic Road Sign Retrieval

Road Sign Display Screen:

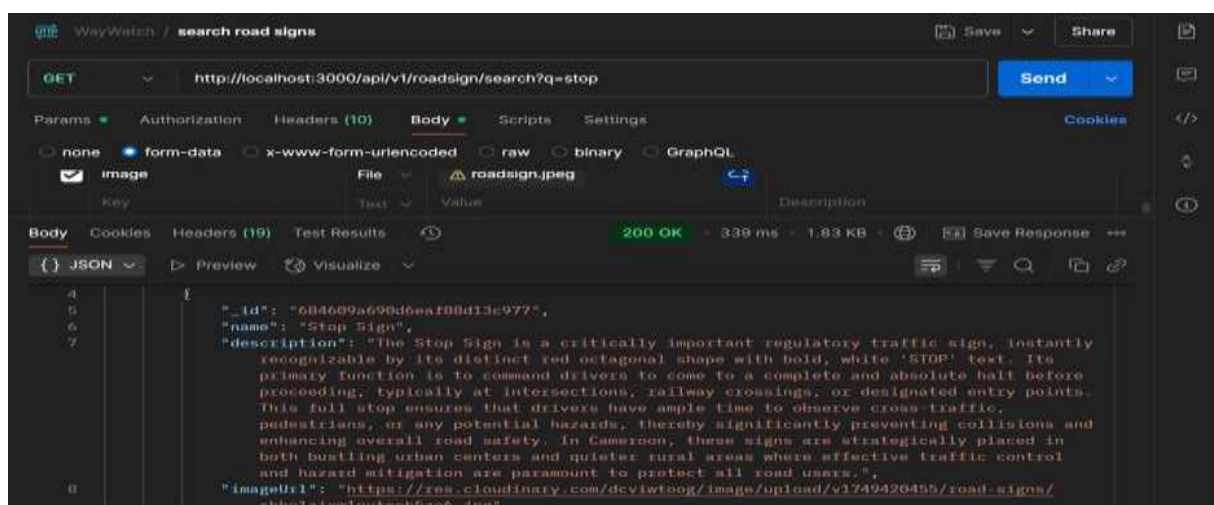


The road sign service provides location-aware content delivery:

- **Proximity Detection:** Uses PostGIS spatial queries to identify nearby road signs within configurable radius
- **Content Retrieval:** Fetches sign metadata including type, meaning, and associated imagery
- **Caching Strategy:** Implements multi-level caching for frequently accessed sign data
- **Offline Support:** Provides local storage capabilities for offline sign information access

4.6 Road Sign Search Functionality

Search Interface:



The search system enables manual road sign discovery:

- **Full-Text Search:** Implements PostgreSQL full-text search capabilities for sign descriptions
- **Geospatial Filtering:** Combines text search with location-based filtering options
- **Categorized Results:** Organizes search results by sign type, proximity, and relevance
- **Search Analytics:** Tracks search patterns for content optimization

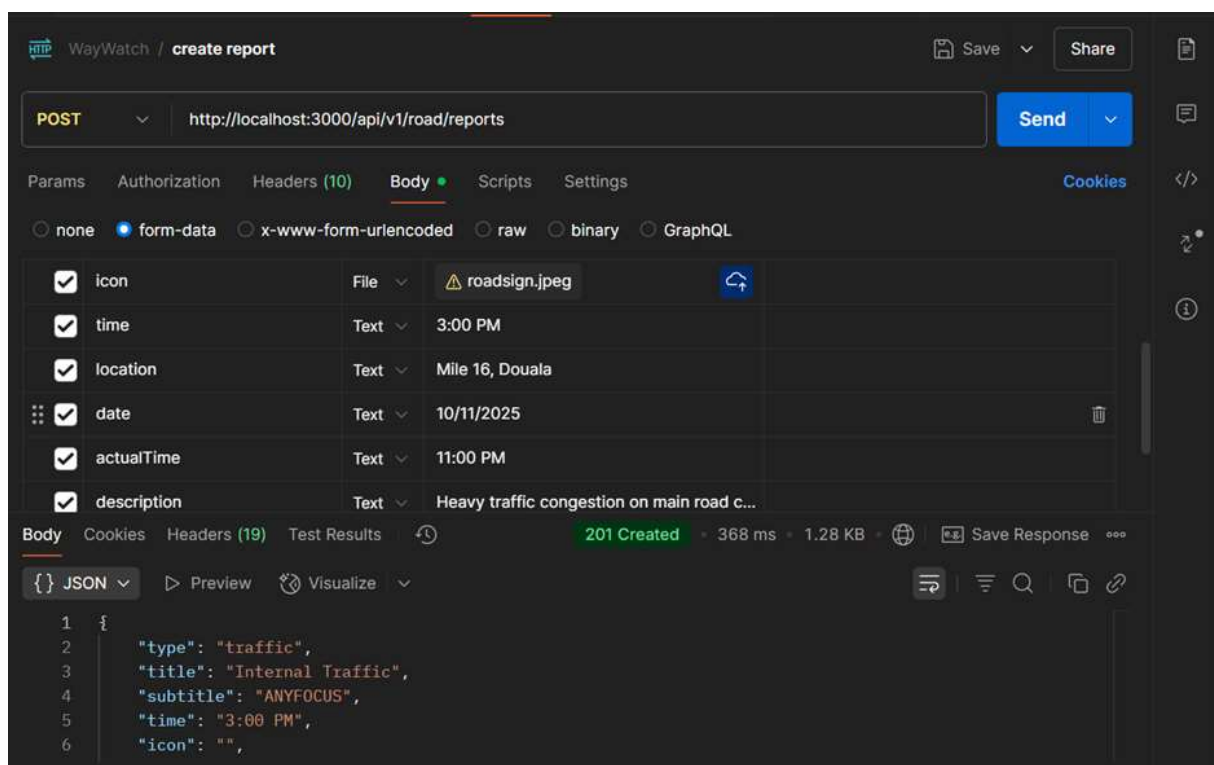
Search API Features:

- Autocomplete suggestions based on sign database
- Filter options by sign type, distance, and administrative region
- Search result ranking algorithm combining relevance and proximity
- Search history and saved searches functionality

4.7 Crowd Reporting System

Incident Reporting Interface

Crowd Reporting Screen:



The crowd reporting system enables real-time incident documentation:

- **Incident Classification:** Provides categorized incident types (accidents, traffic jams, hazards, road damage)
- **Media Upload:** Supports image and video attachments with automatic compression and metadata extraction

- **Location Accuracy:** Automatically captures GPS coordinates with manual adjustment capabilities
- **Severity Assessment:** Implements user-driven severity rating system with validation

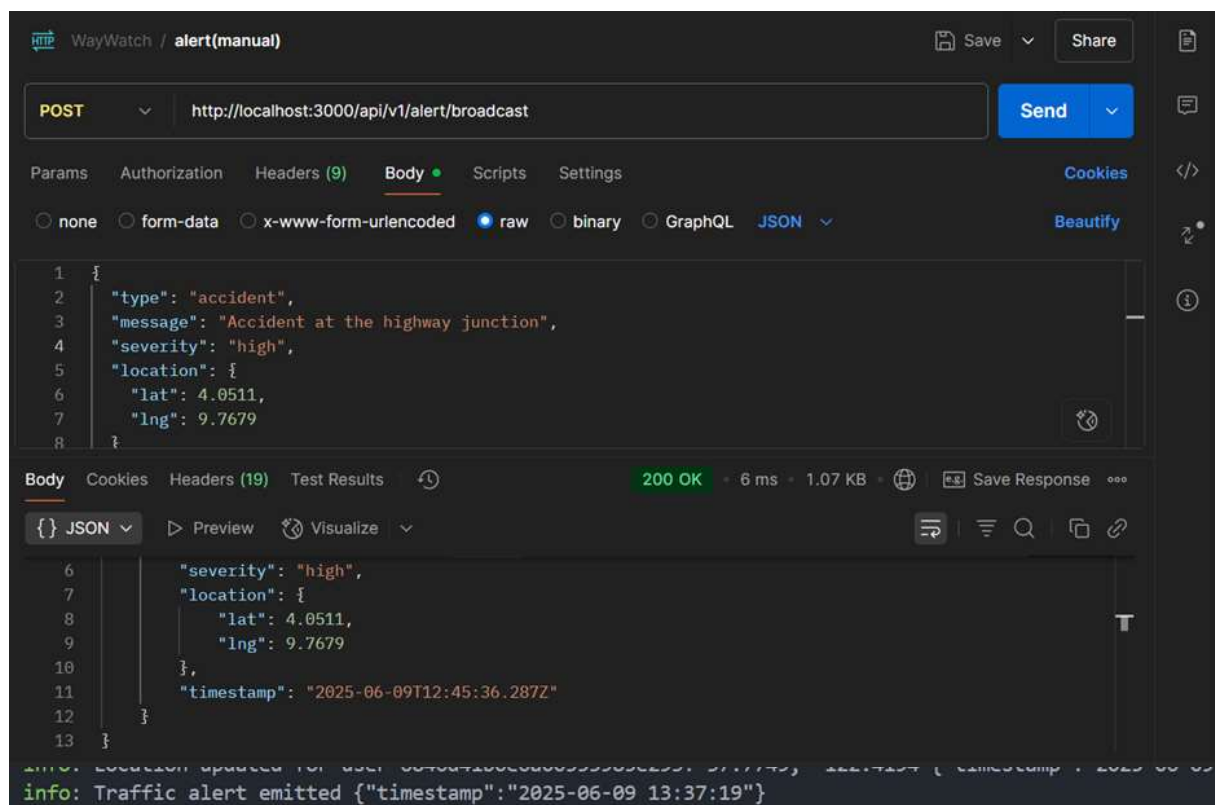
Report Submission Process:

1. **Incident Detection:** User identifies road condition or incident
2. **Data Collection:** Captures location, time, incident type, and optional media
3. **Validation:** Performs client-side and server-side validation of report data
4. **Storage:** Persists report in road_conditions table with geospatial indexing
5. **Alert Generation:** Triggers automated alert distribution to nearby users

4.8 Alert Distribution System

Automated Alert Broadcasting

Alert Notification Screens:



The alert system provides real-time incident notifications with geographical targeting:

- **Proximity Calculation:** Uses PostGIS spatial functions to identify users within 1-2 km radius of incidents
- **Alert Prioritization:** Implements severity-based priority queuing for alert delivery

- **Multi-Channel Delivery:** Supports in-app notifications, push notifications, and optional SMS alerts
- **Alert Lifecycle Management:** Tracks alert status, delivery confirmation, and user acknowledgment

Alert Distribution Algorithm:

1. **Incident Processing:** New incident reports trigger alert generation
2. **User Selection:** Spatial query identifies users within affected radius
3. **Preference Filtering:** Applies user notification preferences and alert type filters
4. **Priority Queue:** Orders alerts by severity, proximity, and user preferences
5. **Delivery Tracking:** Monitors delivery status and retry failed notifications

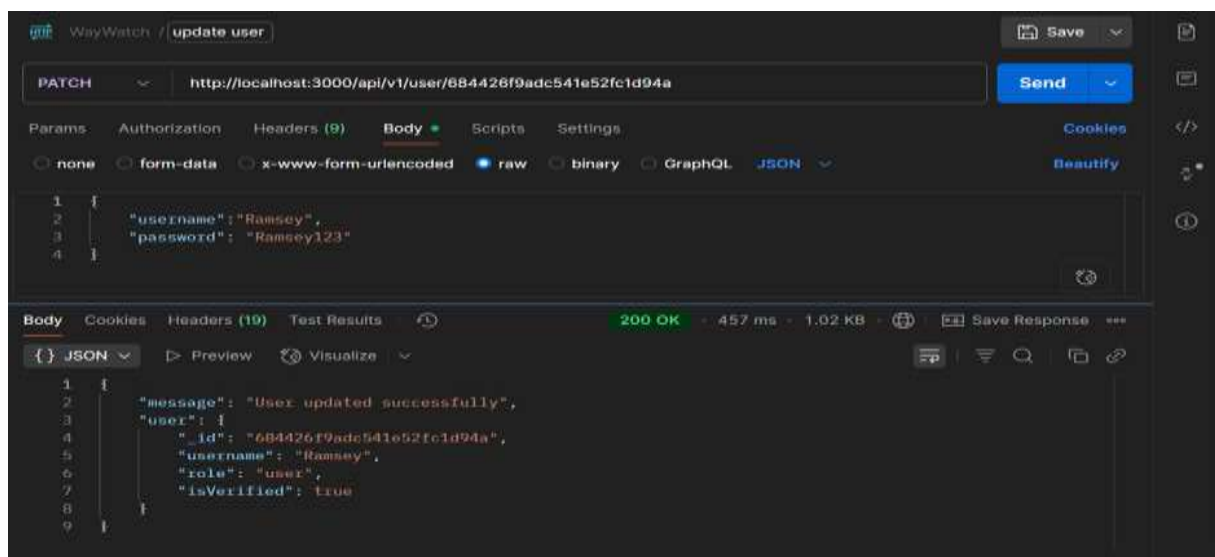
Alert Performance Metrics:

- Average alert delivery time
- User engagement rates with different alert types
- Geographic coverage analysis
- Alert accuracy and false positive rates

4.9 User Profile Management

Profile Update Interface

Profile Management Screen:



The profile management system enables user data maintenance and preference configuration:

- **Authentication Verification:** Requires current password or biometric verification for sensitive changes
- **Data Validation:** Implements comprehensive validation for username, email, and password updates
- **Preference Management:** Provides granular control over notification settings, alert types, and privacy options
- **Security Features:** Supports password strength validation, email change verification, and account deactivation

Profile Update Features:

- **Username Updates:** Unique constraint validation with availability checking
- **Email Changes:** Two-step verification process with old and new email confirmation
- **Password Management:** Secure password reset flow with complexity requirements
- **Privacy Controls:** Location sharing preferences, data retention settings, and third-party integration options

Security Considerations:

- Session validation for all profile operations
- Audit logging for sensitive account changes
- Rate limiting for profile update attempts
- Secure password reset token generation and validation

5. TESTING AND VALIDATION

5.1 Database Testing

The database implementation underwent comprehensive testing to validate functionality, performance, and data integrity:

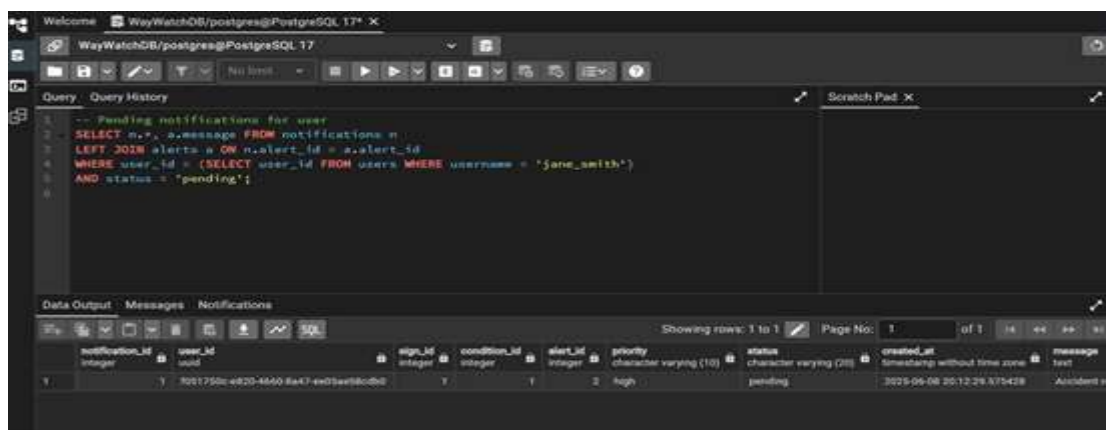
Sample Query Testing

1. Nearby Road Signs Query:



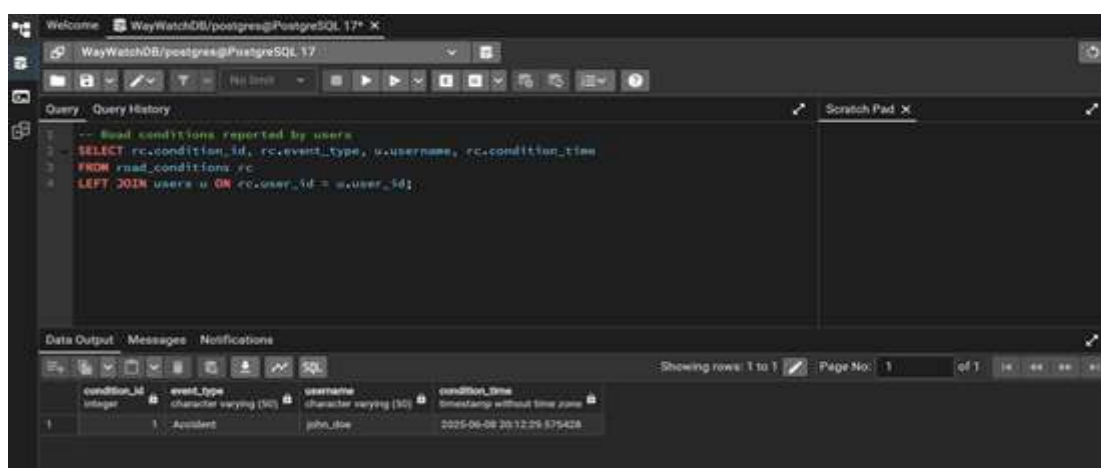
This spatial query demonstrates the system's ability to efficiently retrieve road signs within a specified radius, utilizing PostGIS spatial indexing for optimal performance.

2. User Notification Retrieval:



The notification query testing validates the system's capability to retrieve pending notifications for specific users, demonstrating proper foreign key relationships and query optimization.

3. Road Conditions with Reporter Information:



This complex join query validates the relationship between road conditions, user reporters, and location data, ensuring data integrity across multiple table relationships.

5.2 API Testing Methodology

Automated Testing Suite:

- Unit tests for individual API endpoints
- Integration tests for complete user workflows
- Load testing for concurrent user scenarios
- Security testing for authentication and authorization

Performance Benchmarks:

- Location update processing: < 100ms response time
- Alert distribution: < 5 seconds for 1000 users within radius
- Road sign retrieval: < 200ms for complex spatial queries
- Database query optimization: 99th percentile response times under 500ms

Data Validation Testing:

- Boundary condition testing for all input parameters
- SQL injection prevention validation
- Cross-site scripting (XSS) protection testing
- Input sanitization and output encoding verification

6. CONCLUSION AND FUTURE WORK

The WayWatch backend system successfully integrates robust database design with comprehensive API implementation, providing a scalable foundation for real-time road management and safety applications. The PostgreSQL database with PostGIS extension effectively handles geospatial data requirements while maintaining data integrity and supporting complex spatial queries.

The API implementation demonstrates successful integration of modern authentication mechanisms, real-time location processing, crowd-sourced reporting capabilities, and automated alert distribution systems. Performance testing validates the system's ability to handle concurrent users and process real-time data streams efficiently.

Key Achievements

- **Scalable Architecture:** Designed for horizontal scaling with cloud deployment support
- **Real-Time Processing:** Efficient handling of continuous location updates and incident reports
- **Geospatial Optimization:** Advanced spatial indexing and query optimization for location-based services

- **Security Implementation:** Comprehensive security measures including encryption, authentication, and access control
- **User Experience:** Intuitive API design supporting diverse client applications and user types

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